

What we did

RL on #230's `<contacts-v1.multi>` checkpoint, with the reward computed over the **sections of a single rollout** rather than over the rollouts of a group.

Five reward designs, ~25 scored checkpoints, all on 8 × A100 via SkyRL: **M-C** each section's contribution to its own rollout's consensus · **M-F** the last section's F1 · **M-B** the best section's F1 (oracle) · **M-BC** best + consensus · **M-FC** last + consensus · **M-0** a zero-LR control.

Why

#208 ran eleven runs across five rewards and none improved consensus R-precision. The mechanism was a **unit mismatch**: the reward scored one rollout, while the metric votes over 100 **independent** rollouts — an object no rollout can see.

Under `<contacts-v1.multi>` one rollout emits ~22 contact sets in a single generation, so a reward on those is computed on the kind of object the metric scores, with credit assignment inside the sequence.

The result

****Three arms beat the warm start on every mode**, all CIs excluding zero. Best consensus ****0.5775**** (M-B), best oracle ****0.5663**** (M-B), best final section ****0.5267**** (M-C).**

****The primary criterion is not met.**** 22 independent plain rollouts read 0.5896. At a larger budget the two draw level — 0.6054 against plain-100's 0.6058. So RL ****closed the gap between the multi format and ordinary sampling; it did not open one.****

#208's negative result is a dose-response, not a verdict

Consensus against distance moved: 0.5673 at KL 0, **0.5775 at 0.009**, 0.5529 at 0.031, 0.3969 at 0.486. Every reward helps at small KL and damages at large.

#208 ran its arms at KL 0.06–0.10 and to 3.96 — past the peak on all of them — and its two arms under 0.0015 never moved. **The window it needed lay between the two learning rates it tried.** Two runs at a 3.3× different rate agree to 0.002 at matched KL, so this is distance, not schedule.

Reward design, in three parts

**** $E[r] = 0$ is necessary and not sufficient.**** M-C's per-section marginal is ****366x** larger at 1 section than at 22****** — it paid the policy to emit a worse answer — while $E[A] = 0$ held exactly, because centring is computed **inside** the quantity being gamed.

****Every arm moved its candidate count in the direction its own reward pointed.**** M-C's gradient is strongly negative (collapsed to 1.1 sections); M-B's strongly positive and self-paying (grew to ~26 and held); M-F's nearly flat — and ****a weak gradient is not a safe one, it is an unconstrained one****: M-F ran to 259 sections carrying 1.4 contacts each.

****Gates written against the failure you have seen will miss the next one.**** All three original criteria are one-sided; M-F's failure pushed every one **away** from its threshold and none fired.

Two results that outlive the experiment

****Selection is dominated by aggregation.**** An ORACLE selector of one draft reads 0.5646 where simply voting the same drafts reads 0.5750. A final section should synthesise its predecessors, not pick among them.

****The spread is the resource.**** Making candidates more uniform raises mean section F1 from 0.432 to 0.532 and **lowers** best-of-22 — the same trade that defeats every sharpening reward here, appearing in the corpus's own size law.

What next

****Fix M-C's scale bug and re-run it**** — it is the best final-section arm **despite** the bug. The fix is a rollout-level ``GRPO(C_i(all))`` base (measured scale-correct) plus a ****zero-sum**** shaping term. The obvious repair — scoring against the causal prefix — was tested and ****refuted****.

****Nothing further on M-B.**** Two rates, 13 checkpoints, a smooth peak, agreement to 0.002.

****The diversity gap is a corpus question.**** One rollout's sections cover 658 distinct pairs against 1,065 for 22 independent rollouts.

Figures

`dose\_response.png` — 20 checkpoints, five arms, R-precision against distance.

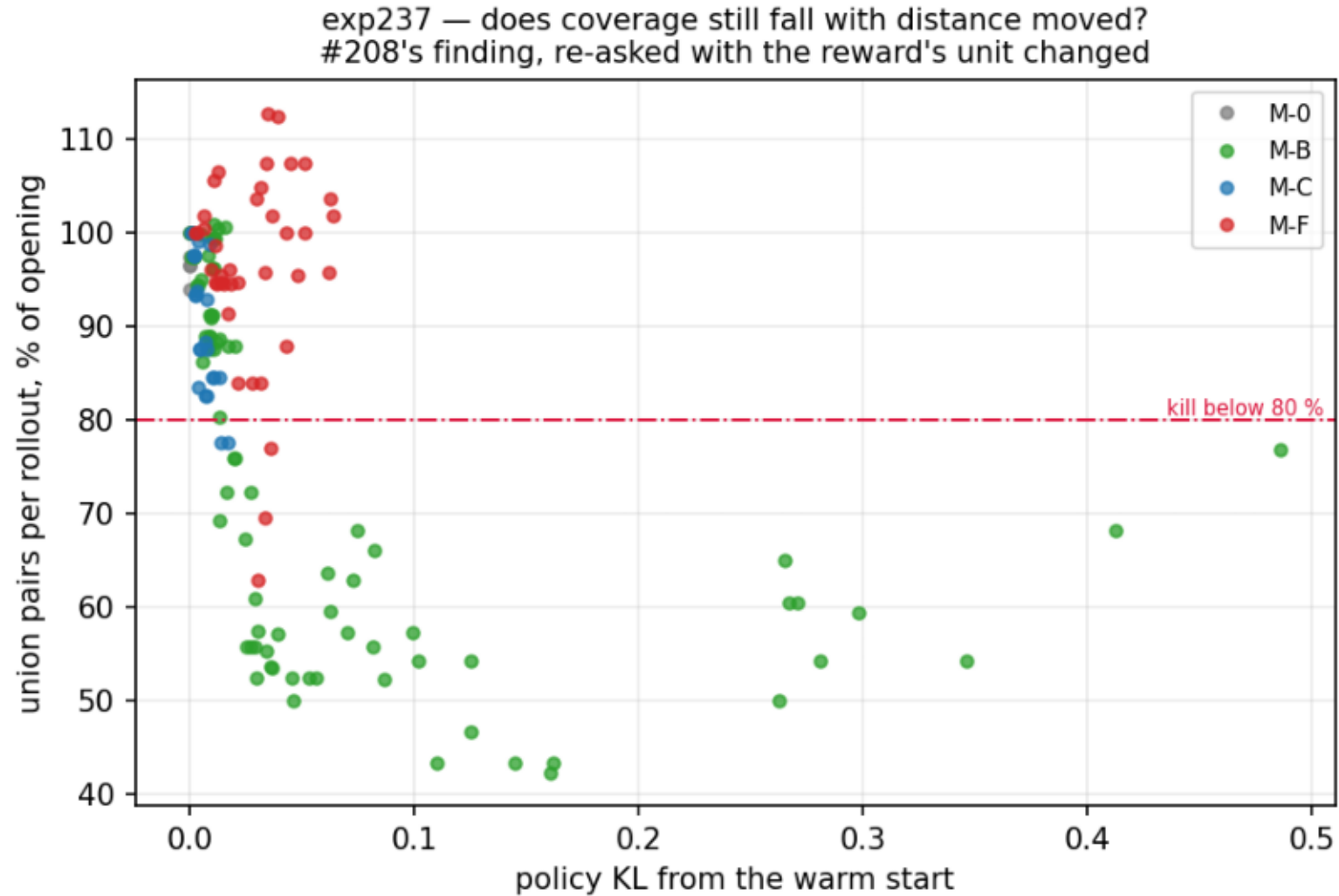
`reward\_vs\_section\_count.png` — each reward's gradient in the count direction.

`section\_count\_incentive.png` — M-C's scale pathology, 366x at one section.

`gates\_over\_training.png` · `reward\_curves.png` · `reward\_shape.png` ·
`coverage\_vs\_kl.png`

coverage_vs_kl.png

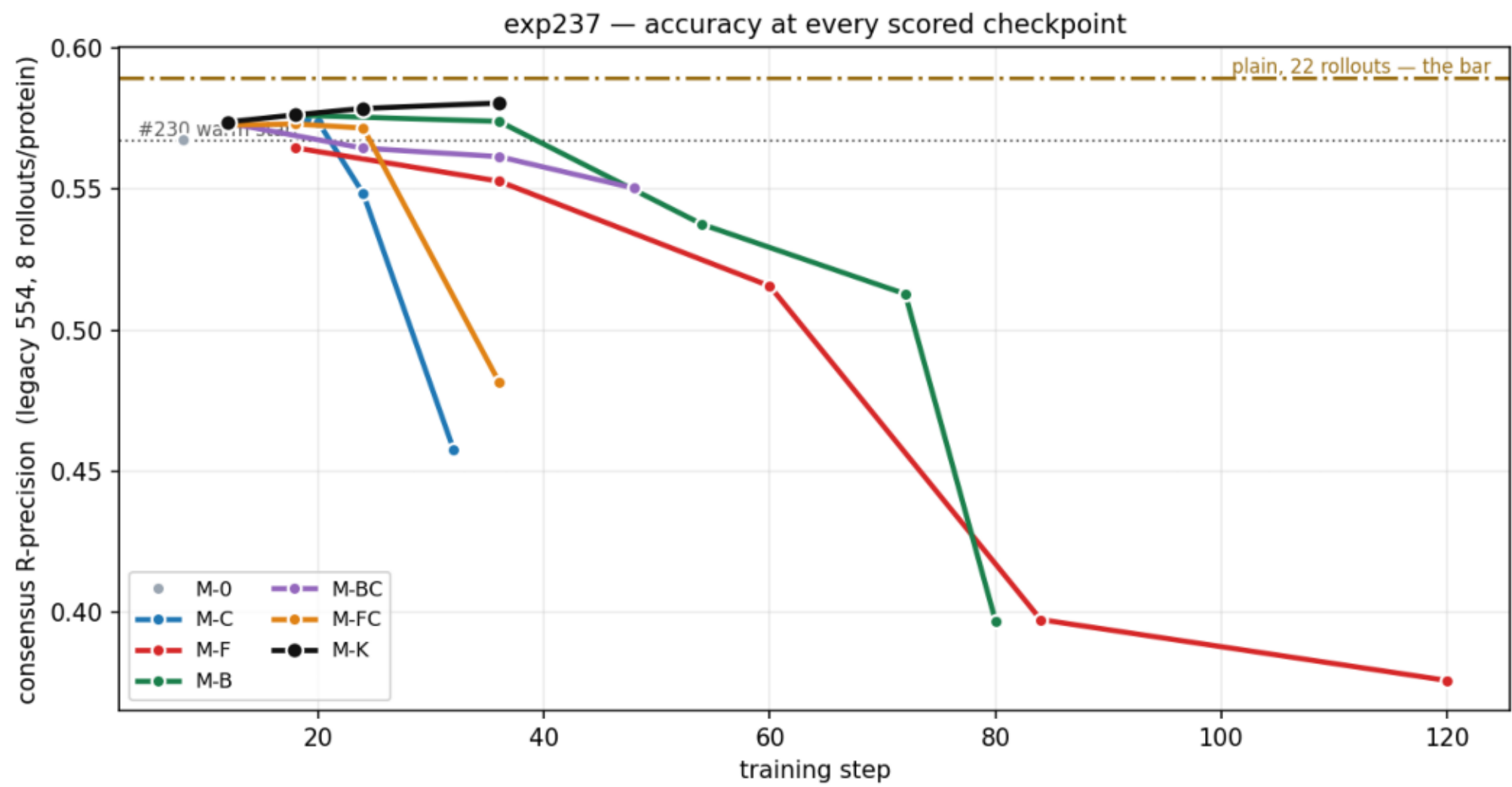
Union coverage against distance moved. M-C crosses the 80% floor by KL 0.013, M-B by 0.016, M-F not until 0.036 -- same destination, different cost per unit of KL.



```
$ make_plots.py --steps data/training_steps.csv.gz --out plots/
```

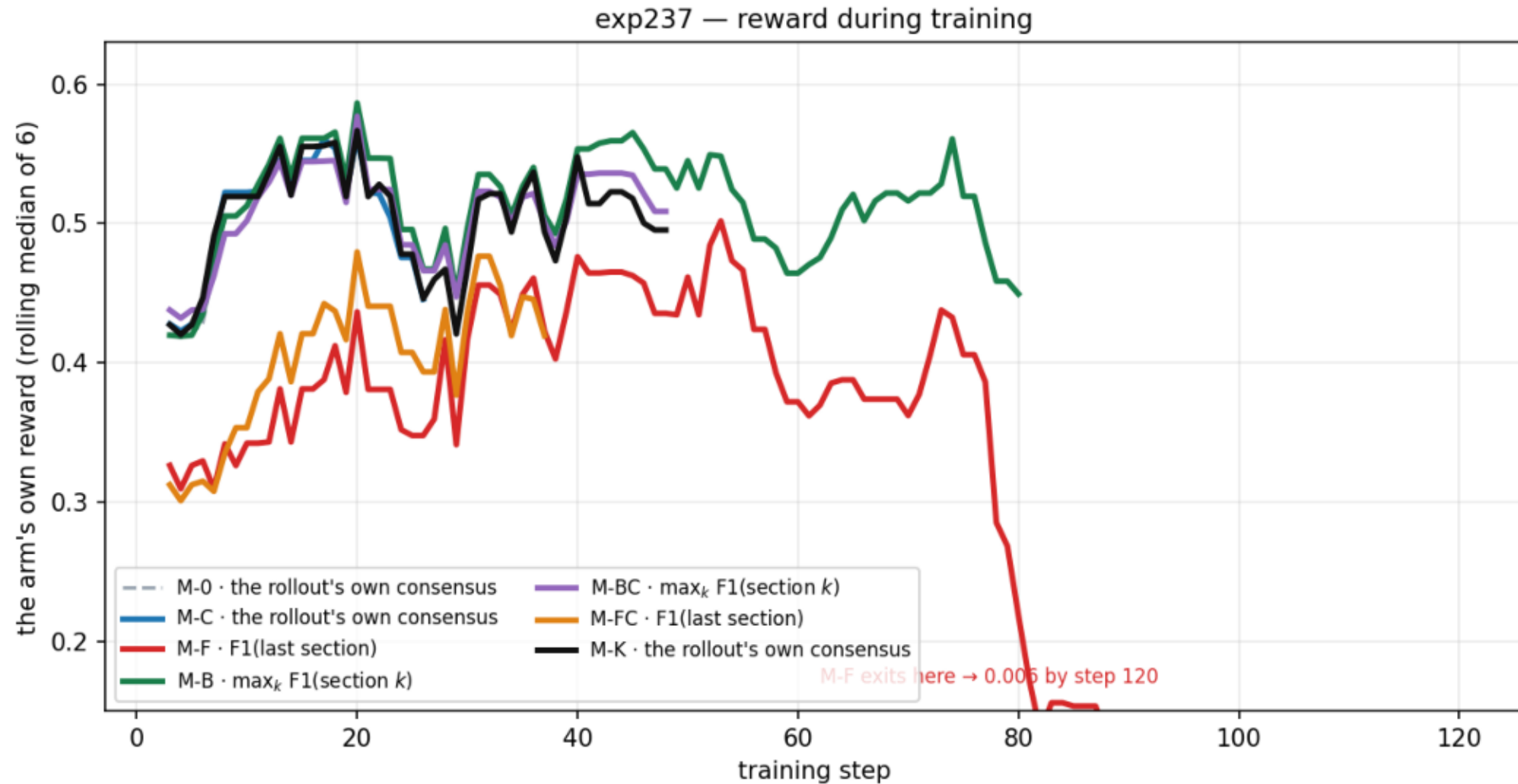
curves_accuracy.png

Consensus R-precision on the 554-protein exp89 benchmark at every checkpoint that was scored. M-K is the only arm still rising at its last scored point.



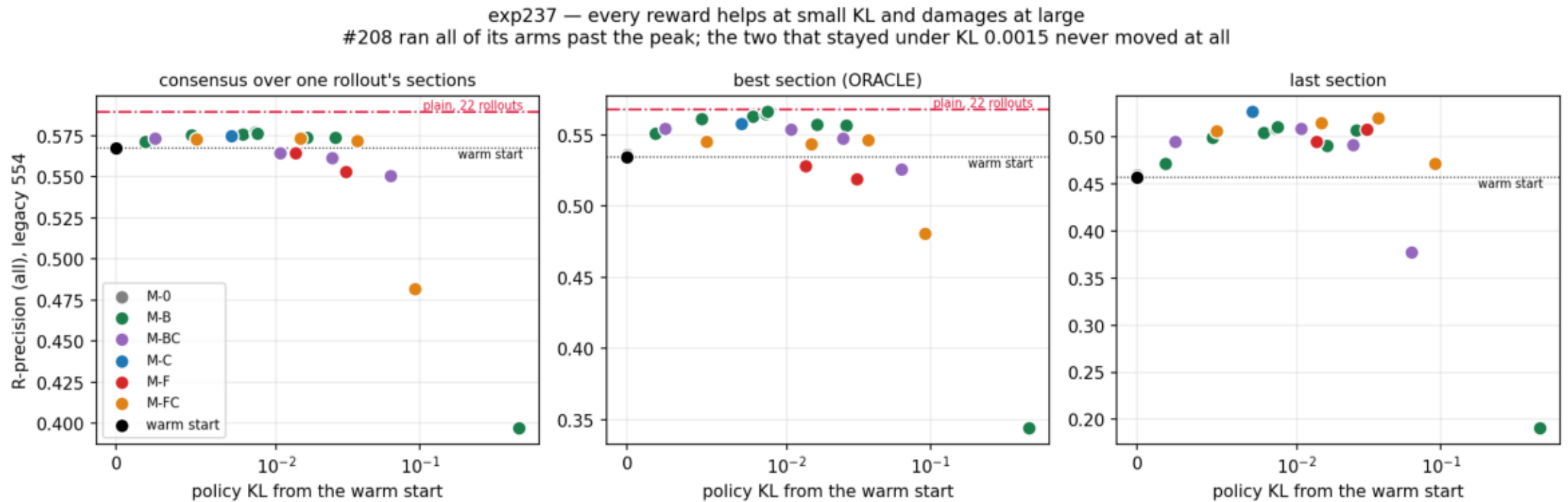
curves_reward.png

Each arm against its OWN reward — the arms do not share one. Rolling median of 6 training batches; the raw per-batch series is dominated by the protein draw.



dose_response.png

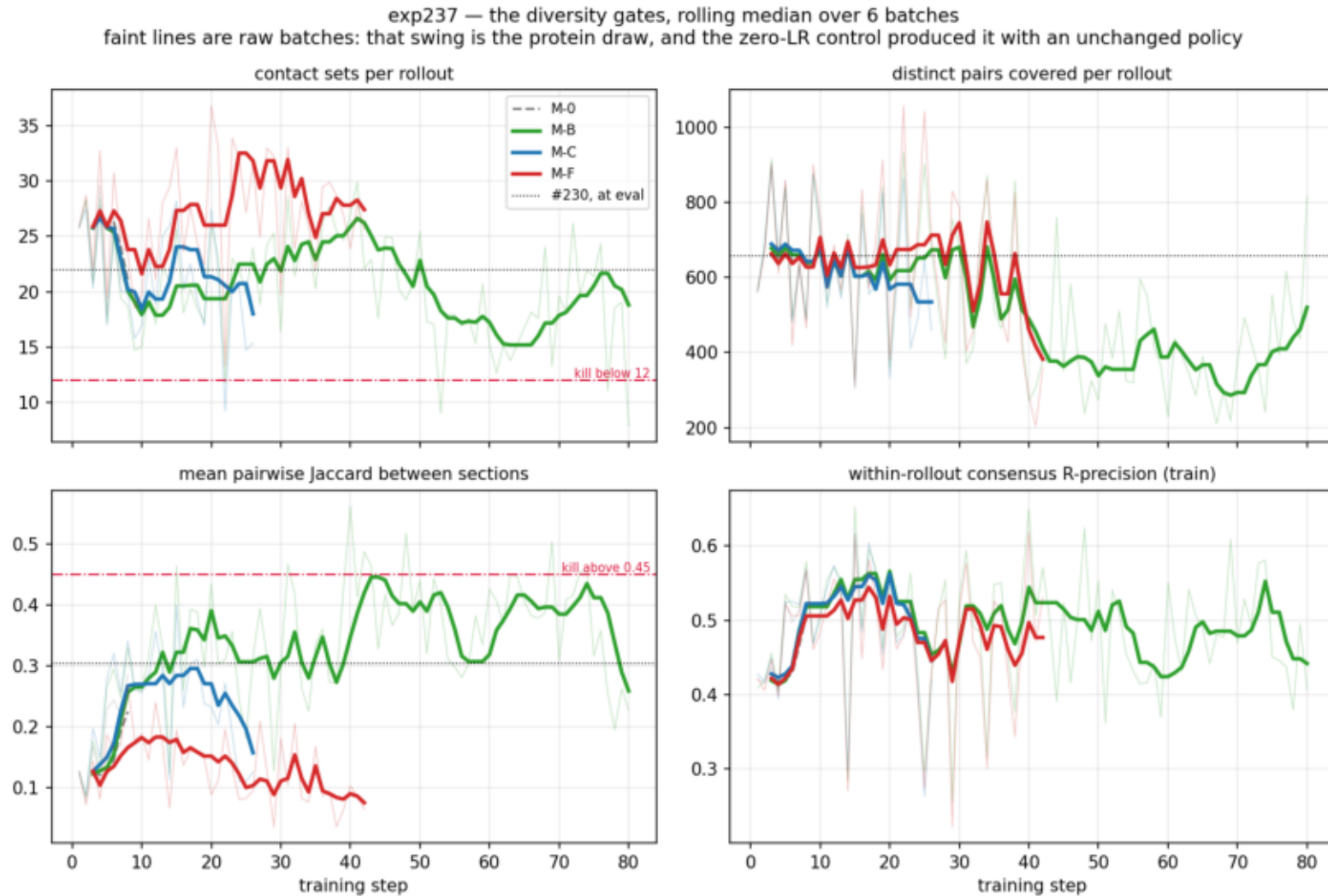
R-precision against distance moved. Consensus peaks at KL 0.007-0.016; #208 ran every arm past it. Red = the budget-matched bar, which nothing reaches.



```
$ plot_reward_shape.py --data data --out plots
```

gates_over_training.png

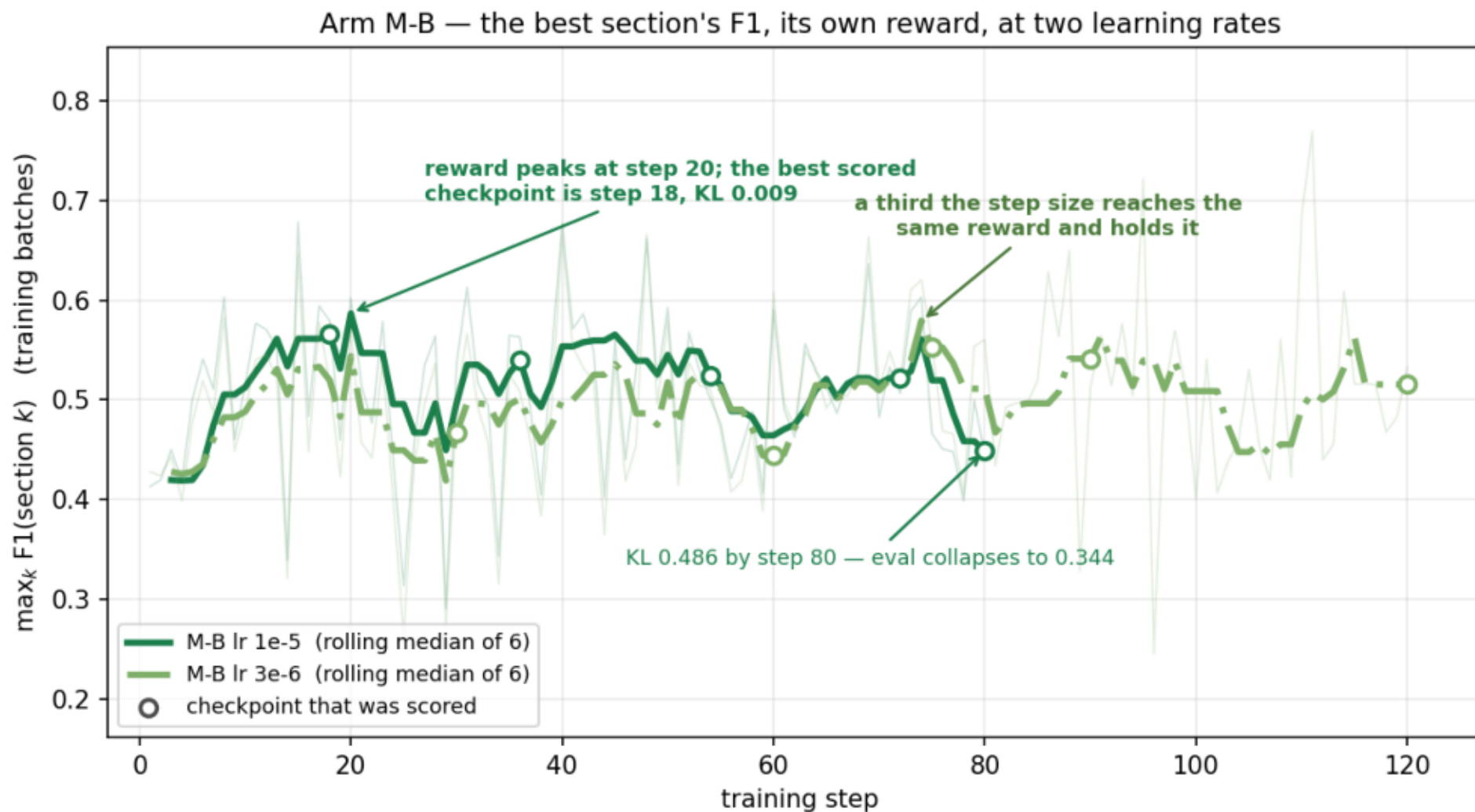
The diversity gates per batch, rolling median over 6. Jaccard separates the arms: M-B rises, M-C and M-F fall. Faint = raw batches, i.e. the protein draw.



```
$ make_plots.py --steps data/training_steps.csv.gz --out plots/
```

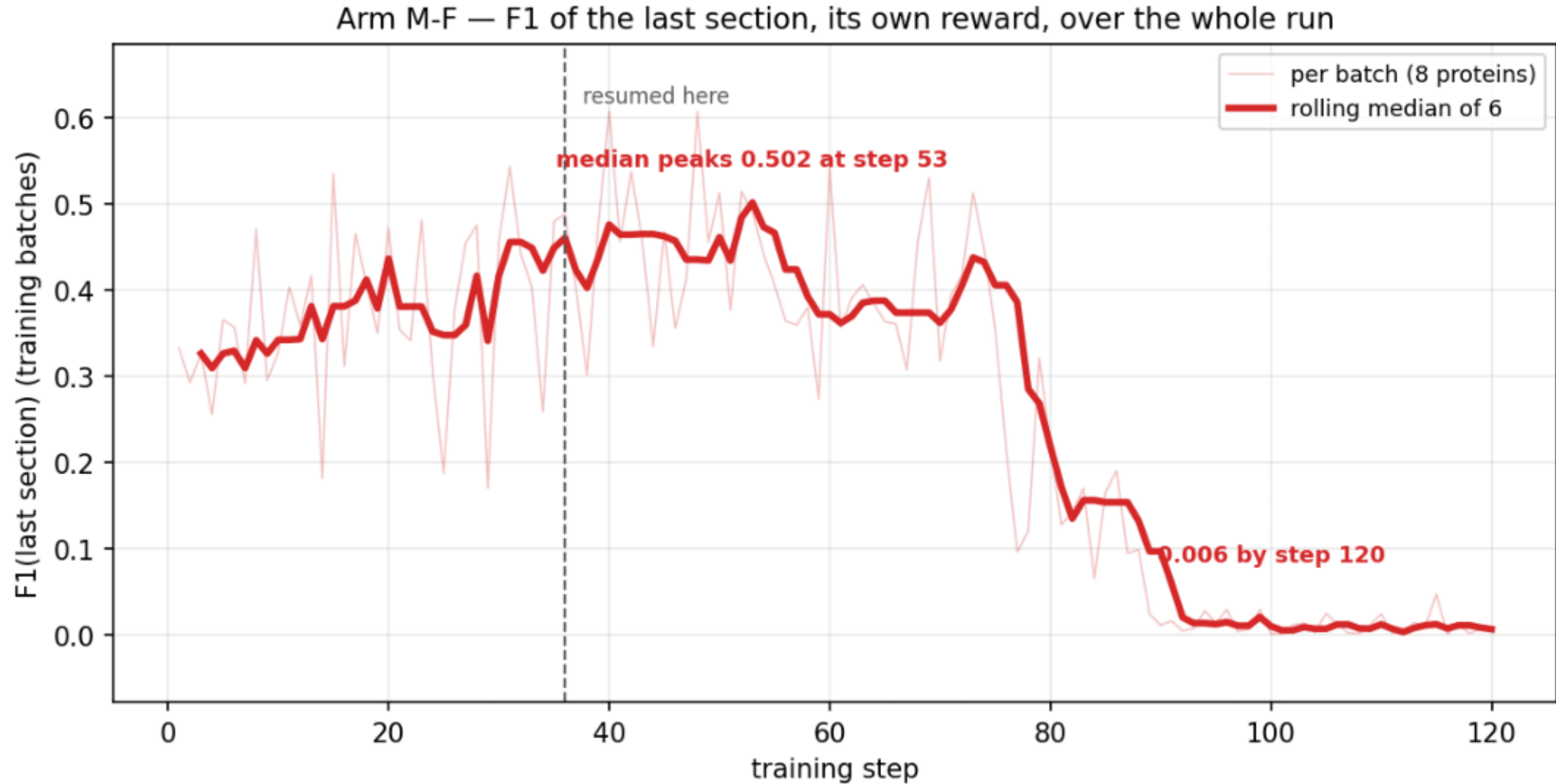
reward_curve_m_b.png

M-B's own reward at lr 1e-5 and 3e-6. Both peak near 0.58; the slow walk holds it while the fast one runs to KL 0.486 and its eval collapses to 0.344. Open circles mark the checkpoints that were scored.



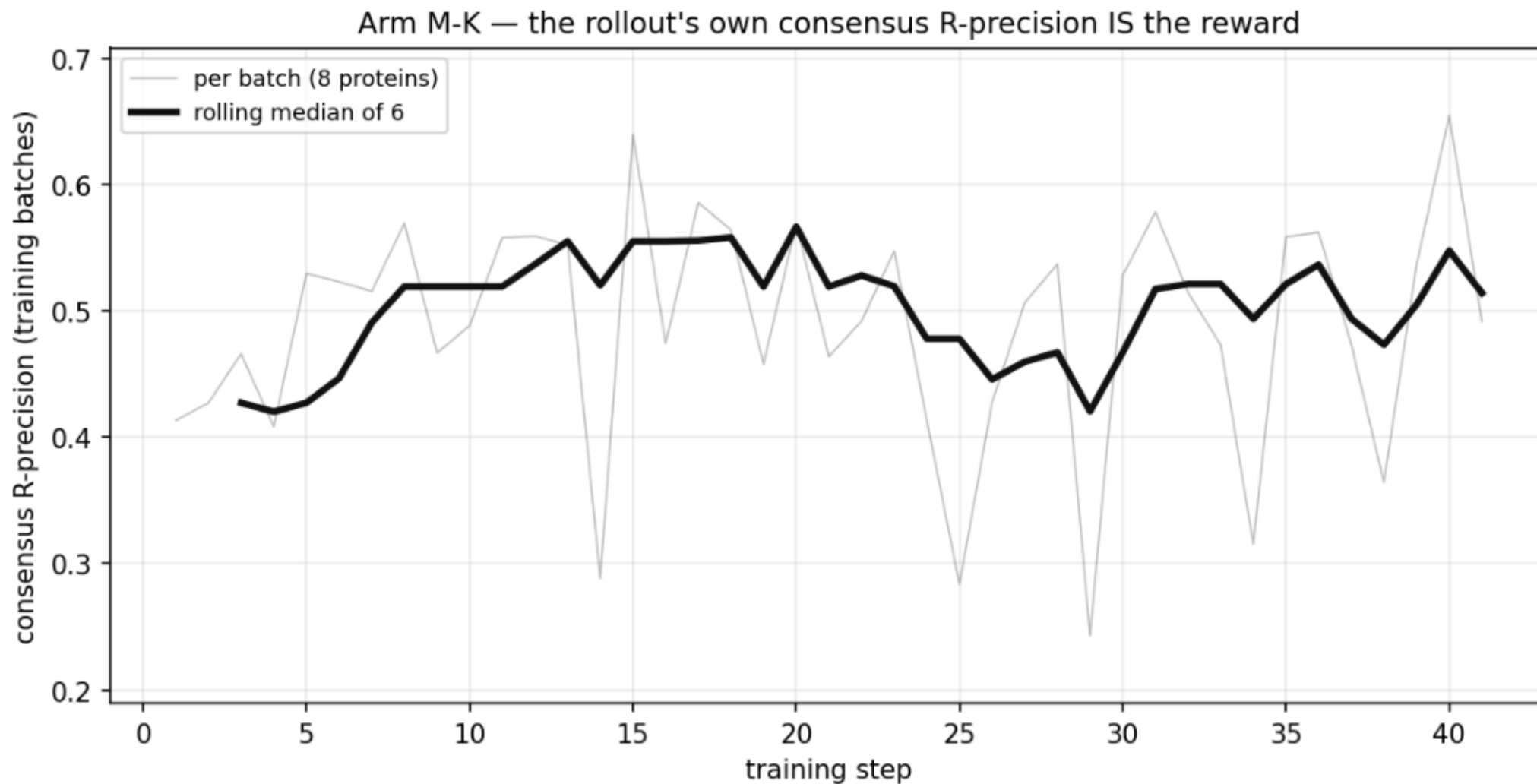
reward_curve_m_f.png

M-F's reward over both runs. It peaks at step 48 and is at 0.006 by step 120: the policy found the unconstrained direction (259 sections of 1.4 contacts each) and took the reward down with it.



reward_curve_m_k.png

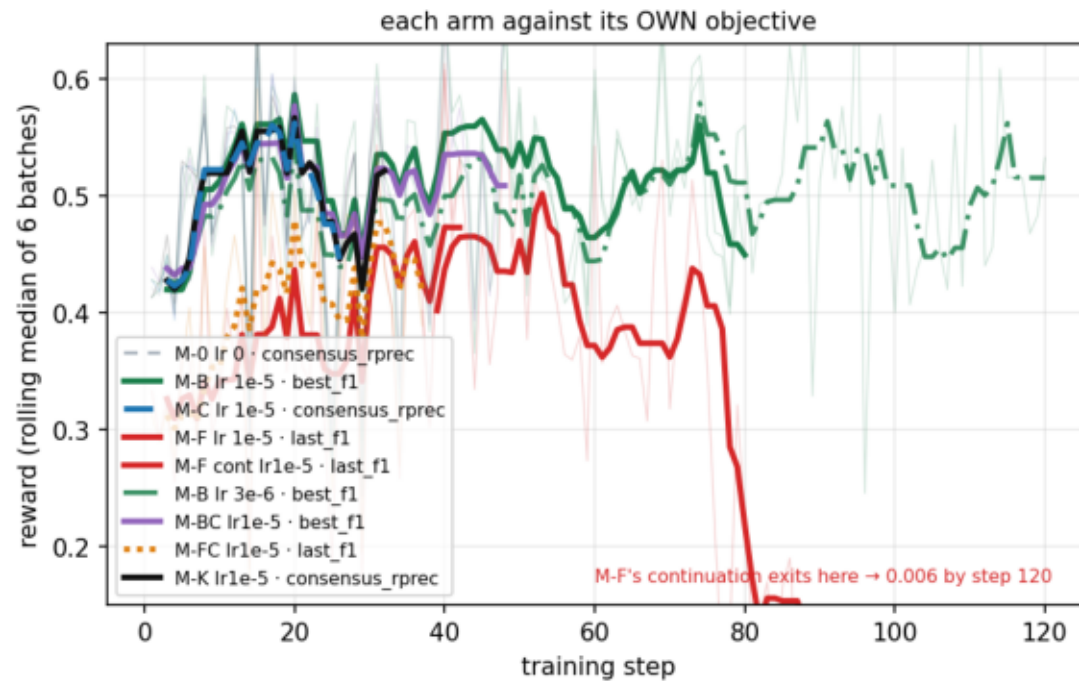
M-K's reward is exactly the quantity plotted: the rollout's own consensus R-precision, one scalar per rollout with a GRPO baseline.



reward_curves.png

Each arm against its own reward (left) and the shared consensus axis (right), rolling median of 6 batches. Training-batch values on 8 proteins, not eval.

exp237 — reward over training. Faint lines are raw batches: that swing is the protein draw, and the lr-0 control produced it with a policy that never changed.

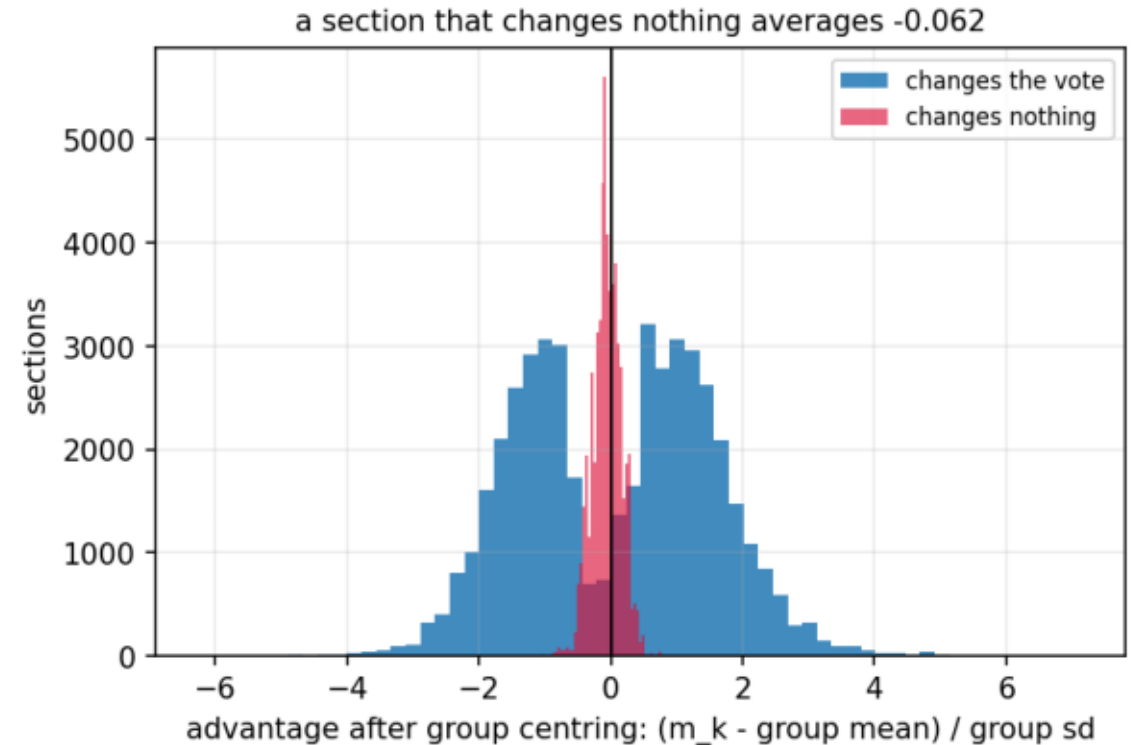
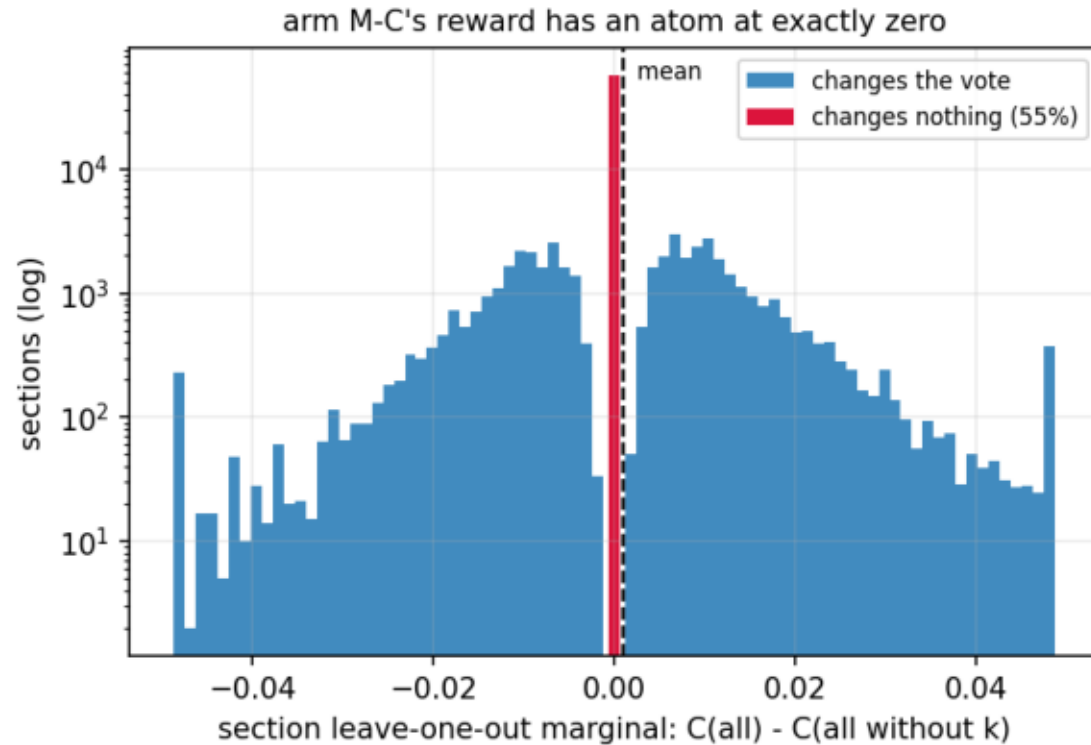


```
$ plot_reward_curves.py --csv data/training_steps.csv.gz --run M-F cont lr1e-5=/tmp/claude-1000/logs_final/exp237_m_f_lr1e-5.log --offset M-F cont lr1e-5=36 --run M-B lr3e-6=/tmp/claude-1000
```

reward_shape.png

Arm M-C's reward before training: 55% of sections change no vote, and average -0.062 once centred. M-F has no such atom and collapsed the same -- so this is not the cause. See RESULTS.md.

exp237 — why $E[A] = 0$ holding exactly was not enough

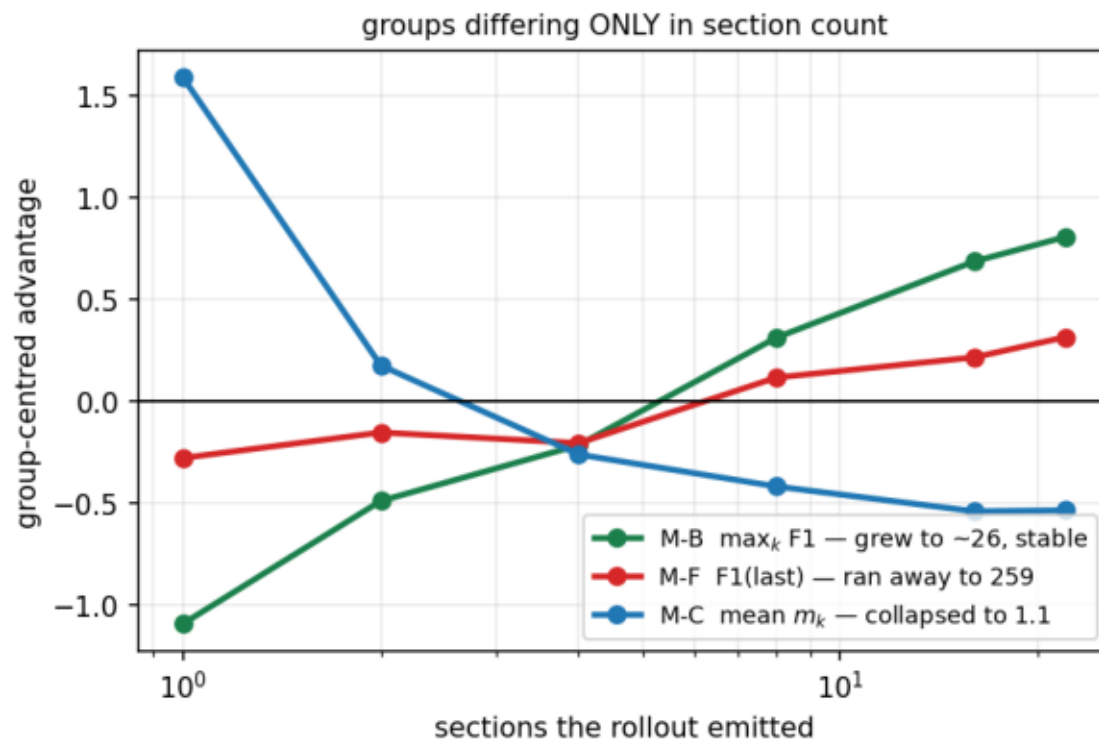
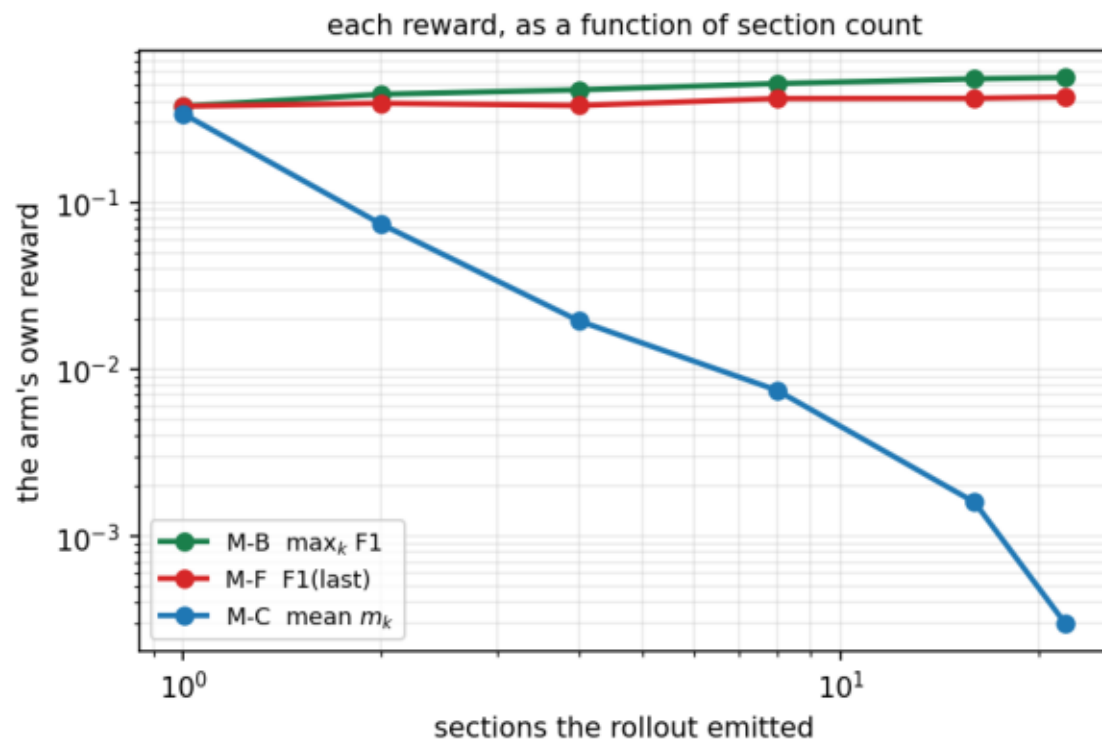


```
$ plot_reward_shape.py --data data --out plots
```

reward_vs_section_count.png

Each reward's gradient in the section-count direction predicts what that arm did: M-C falls (collapsed to 1.1), M-B rises (grew to ~26), M-F is nearly flat (unconstrained, ran to 259).

exp237 — every arm moved its section count in the direction its own reward pointed



section_count_incentive.png

m_k is not scale-free in the section count: at 1 section it is 366x its value at 22, while the rollout's own consensus falls 0.543 -> 0.341. Centring does not remove it -- the shorter rollout is the one above its group's mean.

exp237 — arm M-C's reward pays a rollout for emitting fewer sections

